MILITARY SPECIFICATION

TRUCK, TANK A/S32A-24

This specification is approved for use within the department of the Air Force and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 250-gallon tank truck capable of handling individually 250 gallons of potable water, demineralized water, or water/alcohol mixtures.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

ZZ-H-561	Hose, Rubber and Hose Assemblies, Rubber, Smooth Bore, Water
ZZ-H-601	Hose and Hose Assemblies, Rubber (Wrapped of Braided) Water Service
O-E-760	Ethyl Alcohol (Ethanol), Denatured Alchol, Proprietary Solvents and Special Industrial Solvents
VV-F-800	Fuel Oil, Diesel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMEDTA, Robins AFB, GA 31098-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-C-3432	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-C-4339	Corrosion Preventative, Soluble Oil, for Water Injection Systems
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-C-10387	Coupling, Clamp, Pipe with Bolts and Synthetic Rubber Gasket for Grooved Pipe and Tube
MIL-D-16791	Detergent, General Purpose (Liquid, Nonionic)
MIL-C-27487	Coupling Halves, Quick Disconnect, Cam Locking Type
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-E-52798	Enamel, Alkyd, Camouflage
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-R-83232/2	Reel, Static Discharge, Grounding RMU-25/E, 50 Foot with Connector
MIL-R-83232/3	Reel Static Discharge, Grounding RMU-26/E, 50 Foot with Plug
MIL-C-83286	Coating, Urethane Aliphatic Isocyanate, for Aerospace Applications
MIL-W-83420	Wire Rope, Flexible, For Aircraft Control
MIL-F-83428	Filter Element, Fluid, Pressure MXE-649/E

STANDARDS

Federal

FED-STD-307	Trucks, Light Commercial, Gasoline Diesel Fueled, Two Wheel Driven (4X2 3,000 TO 11,000 GVW)
FED-STD-595	Colors

MILITARY

MIL-STD-12	Abbreviations for Use on Drawings and on Specifications, Standards, and Technical Documents
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program for Systems and Equipment
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-808	Finishes, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1223	Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards

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MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1595	Aerospace Welder Performance Qualifications
MIL-STD-1971	Designing for Internal Aeiral Delivery in Fixed Wing Aircraft.
MS-3506	Connector, Receptacle, External Electric Power, Aircraft, 28 Volt, DC Operating Power
MS-21391	Coupling End, One Inch, Drinking Water, Service, Aircraft, Standard Dimensions for

2.1.2 Other Government Documents, Drawings and Publications. The following other government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitaion.

Other Government documents

AFSC DH 1-11 Air Transportability

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION
Occupational Safety and Health Act (OSHA) with Supplemental Revisions

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services

Copies of specifications, standards, handbooks, drawings, publications, and other government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DODISS shall be the issue of the non-goverment documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

S1.4 General Purpose Sound Level Meters
(Applications for copies should be addressed to the American National Standards Institute, 10 East 40th Street, New York, NY 10016).

SOCIETY OF AUTOMOTIVE ENGINEERS, INCORPORATED

J534 Lubrication Fittings
J682 Rear Wheel Splash and Stone Throw Protection
J1128 Low Tension Primary Cable
J1292 Automobile, Truck, Truck-Tractor, Trailer and Motor Coach Wiring
(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pa 15096).

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NATIONAL FIRE PROTECTION ASSOCIATION

Pamphlet 70 National Electrical Code (Electric Wiring and Apparatus) (Applications for copies should be addressed to the National Fire Protection Association, 60 Battymarch Street, Boston MA 02100).

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Components. The truck shall consist of the following major components:

a. Chassis (see 3.7.1)
b. Tank (see 3.7.19)
c. Pumping compartment (see 3.7.20)
d. Pumping system (see 3.7.21).

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specificed herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type equipment. Except as other wise specified, wood shall not be used.

3.4.1 Toxic products and formulations. Materials used in the construction or treatment of surfaces which may be in contact with potable water during use shall have no adverse effect on the health of the personnel nor impart taste to the water. In addition, these materials shall not deteriorate at such a rapid rate as to limit the useful life of the truck.

3.4.2 Fungus proof materials. Materials which are not nutrients for fungi shall be used to the greatest practical extent. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent acceptable to the procuring activity.

3.4.3 Protective treatment. Materials that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage shall be protected against deterioration in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack or scale with age or extremes of climatic conditions shall not be used.

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3.5 Design and construction. The vehicle shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping and storage with minimum loss of time for maintenance, repair and servicing. The design of the vehicle shall ensure safe, efficient and economical operation. All assemblies, controls and installed equipment shall be located so that there is no adverse interference with each other or with the operation and shall be readily accessible for maintenance, operation, removal and replacement. There shall be no evidence of accelerated wear, failure or permanent deformation when operated as intended.

3.5.1 Functional design. The truck shall be designed to satisfactorily receive, transport, store and service potable or demineralized water. With the incorporation of the meter (see 3.7.21.15) into the pumping system, the truck shall also be capable of satisfactorily receiving and servicing water/alcohol mixtures. The truck shall be designed for world wide operation at fixed bases.

3.5.2 Reliability and Maintainability. The vehicle shall be reliable in accordance with Section 4, and designed and constructed in accordance with MIL-STD-470 paragraph titled Established Maintainability Design Criteria to provide:

a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources and maintaining the vehicle, and a minimum number of tools and equipment for maintenance.

c. Ease of adjusting, servicing and replacing parts and components.

3.5.3 Foolproofness. Where improper installation of an item could cause a malfunction, an unsymmetric mounting system shall be provided to ensure proper mounting of the item.

3.5.4 Fastening devices and foreign object damage. All screws, bolts, nuts, pins, caps and other fastening devices shall be securely attached and shall be designed, manufactured, and installed with means of preventing loss of torque or adjustment. Cotter pins, locks washers or nylon patches shall not be used for this purpose, except on commercial chassis, tapped aluminum threads shall have a minimum thread engagement of at least 2 times the nominal fastener diameter.

3.5.5 Servicing provisions and lubrication. All parts requiring grease for lubrication shall be provided with fittings in accordance with SAE J534. A pressure relief device shall be provided where excessive pressures may damage grease seals. Drains, grease fittings, lines and check points for all components will be accessible and not require special tools for servicing. Extended fittings shall be utilized for nonaccessible parts or assemblies. All servicing provisions and lubrication parts shall be so designed as to allow access by personnel wearing arctic winter clothing.

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3.5.6 Component accessibility and protection. The truck shall be so designed, that all space in which work is performed during operation, service and maintenance of the vehicle shall be free of hazardous protrusions, sharp edges and that all components of the mounted equipment may be readily removed, serviced, and operated by arctic clad personnel. The vehicle shall be designed to insure ease of production, rapid installation of equipment and general maintenance and repair. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized shall be guarded or insulated.

3.5.7 Bushings and bearings. Bushings and bearings, as appropriated, shall be provided at any location where there is relative rotation between two parts, except hinges.

3.5.8 Screw assemblies. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage. All threads shall match the mating threads and be free of burrs.

3.5.8.1 Screw threads. Screw threads shall conform to Handbook H 28.

3.5.9 Envinronmental conditions. The vehicle shall be capable of satisfactory storage, start and operation under the following environmental conditions with truck stabilized at:

a. Temperatures ranging from -40°F to +125°F.

b. Exposure to relative humidities up to 100 percent.

c. Exposure to salt fog.

d. Exposure to sand and dust particles as encountered in desert areas.

3.5.10 Riveting. Riveting operations shall be performed to ensure tight fit and complete heading of the rivets.

3.5.11 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for adequate adjustment, gear assemblies shall be free of backlash.

3.5.12 Cleaning. The vehicle shall be thoroughly cleaned. Loose, spattered or excess solder, welding slugs, stray bolts, nuts and washers, rust metal particles, pipe compound and other foreign matter shall be removed during and after final assembly.

3.5.13 Life. The vehicle shall be designed for a life expectancy of at least 10 years.

3.5.14 Weight distribution. The weight distribution of the vehicle in any loading condition shall not be unbalanced or improperly distributed. The difference in load between the left and right wheels of any axle shall not exceed 2.5 percent of the total load on the axle.

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3.5.15 Electromagnetic interference. The vehicle shall comply with the requirements of MIL-STD-461, Class C1, Group II for electromagnetic interference, except that the limits for UM03 (Figure 8-1) shall be relaxed by 20dB.

3.5.16 Air transportability. The vehicle shall be air transportable in accordance with MIL-STD-1791 and AFSC DH 1-11 in Types C-130, C-141 and C-5 Aircraft. Removal of equipment shall be permitted if necessary to provide the required height and width for air transport. Removal and reinstallation of equipment shall be limited to removal and attachment of standard fasteners and fittings with common hand tools. The weight of individual parts or assemblies which must be handled or positioned shall not exceed the limits specified in MIL-STD-1472 for a one person lift. All equipment removed shall be stored on the vehicle; all necessary caps and plugs shall be provided to permit driving and storage of the vehicle while in the air transport configuration. Removal and installation of the equipment shall take no longer than 15 minutes for two mechanics using common hand tools.

3.5.17 Metals.

3.5.17.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5.17.2 Stainless steel. Unless otherwise specified herein, the tank and pumping system and all components which come into contact with water shall be fabricated of 300 series stainless steel.

3.5.18 Bonding. Metal parts shall be specifically bonded together to minimize the possibility of sparks.

3.5.19 Elastomers. Elastomeric materials used in the tank, pumping system and all components which come into contact with water shall be compatible with potable water, demineralized water, water/alcohol mixtures and 50/50 mixture of propylene glycol (uninhibited) antifreeze and demineralized water.

3.5.20 Safety. All equipment furnished under this specification shall be certified as complying with the Department of Labor Standards of the Occupational Safety and Health Administration in effect on the date of manufacture.

3.5.21 Handtools. The truck shall be so designed and constructed that general maintenance, assembly and disassembly can be performed to the maximum practical extent using common handtools. All hinges, latches and frequently replaced hardware shall be bolted, not welded, to the vehicle.

3.5.22 Welders and welding.

3.5.22.1 Welders. All welders employed in the fabrication of the vehicle shall be certified in accordance with the applicable requirements of MIL-STD-1595, the American Welding Society or the American Society of Mechanical Engineers.

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3.5.22.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall transmit stress without cracking or permanent destortion when the parts connected by the welds are subjected to test, proof and service loadings.

3.5.23 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808, except that commercial components may be prepared in accordance with the manufacturer's standard practice, provided it is compatible with the exterior finish paint.

3.5.24 Finish. The exterior paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion, or other defects, and shall be capable of passing the test described in paragraph 4.8.10 of MIL-C-83286.

3.5.24.1 Forest Green. Unless otherwise specified, the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint, Forest Green, Color Number 24052 of FED-STD-595. The paint shall be in accordance with MIL-C-83286, or shall be DuPont Imron, or equal. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, bumpers, light trim rings, door handles and drip rails) shall be painted with the same paint as the exterior of the vehicle.

NOTE

"Equal" is defined as having performance identical to or superior to DuPont Imron when subjected to the following tests:

a. Tests in accordance with MIL-C-46168:

(1) Paragraph 4.3.17 Flexibility.

(2) Paragraph 4.3.23 Accelerated weathering.

b. Tests in accordance with MIL-C-83286:

(1) Paragraph 4.8.6 Low temperature resistance.

(2) Paragraph 4.8.10 Resistance to taping.

(3) Paragraph 4.8.12 Pencil hardness.

3.5.24.2 Desert Sand. When specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30315 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted with the same paint as the exterior of the vehicle.

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3.5.24.3 Rustproofing. Unless otherwise specified (see 6.2), the vehicle shall be rustproofed in accordance with the contractor's standard commercial practice. The first article units shall not be rustproofed until after approval of the first article test report.

3.5.24.4 Markings. Unless otherwise specified, markings for vehicles painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595, and markings for vehicles painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.5.25 Operational markings.

3.5.25.1 Identification markings. Unless otherwise specified (see 6.2), the vehicle shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.5.25.2 Lettering and numbering. All lettering and numbering shall be reflectorized tape as specified in MIL-STD-808 and by the procuring activity.

3.5.25.3 Nameplates. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130. The plates shall be made of a noncorrosive steel.

3.5.25.4 Control markings. All controls, valves, gauges and indicators used in the operation of the vehicle shall be identified by securely attached nameplates of such construction that exposure to fuel, oil, water, dirt, light, etc, will neither fade nor eradicate them. Tags or decals shall not be used. The plates shall be permanently fastened and conspicuously located on the reference assembly.

3.5.25.5 Operating instructions and data plates. All data plates, placards, charts, instruction plates, etc, shall conform to MIL-P-514 and be of a size and shape consistent with the information required in this specification. Abbreviations shall be in accordance with MIL-STD-12.

3.5.25.6 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise and adequate to enable operation of the vehicle without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.5.25.7 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

3.5.25.8 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

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a. Side and rear silhouette views of the vehicle.

b. Horizontal and vertical location of the center-of-gravity of the vehicle in transport configuration.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width and height.

f. Tiedown instruction.

3.5.25.9 Data plates. Instruction and identification data shall be permanently and legibly etched upon metal plates in accordance with MIL-P-514. The instruction plates shall be permanently attached on the inside of the compartment doors so as to be most accessible and visible to the operator standing in front of the control panel. The instruction and other plates shall be aluminum and not less than 0.032 inch thick. All plates shall be anodized or etched to produce raised natural metallic colored markings with a dark color background.

3.5.25.10 Hose reel instruction plate. An instruction plate shall be provided which cautions the operator to control the rewind rate of the hose by holding the nozzle. If the hose is allowed to freely rewind onto the reel, damage could occur to the reel and nozzle. The instruction plate shall be mounted in a conspicuous location near the hose reel.

3.5.25.11 Pump compartment heating system instruction plate. A concise step-by-step operating instruction plate containing procedures to be followed for operating the heating system and precautionary measures to be taken by the operator shall be permanently attached to the dash panel of the truck cab and adjacent to the heater controls (see 3.7.21.33.5).

3.5.25.12 Preproduction plates. Unless otherwise specified by the procuring activity (see 6.2), the preproduction trucks may be equipped with data plates consisting of typed lettering on paper laminated in plastic or similar semipermanent construction. The preproduction trucks shall contain sufficient indentification and other data plates to provide examples of plates to be furnished on production trucks. The composition of all plates shall require the approval of the procuring activity. The preproduction trucks shall incorporate the final production model plates prior to delivery.

3.5.25.13 Truck identification Plate. The truck shall be marked for identification in accordance with MIL-STD-130. The plate shall be permanently attached to the dash panel in the cab of the truck. The plate shall contain at least the following information:

NOMENCLATURE

USAF

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REGISTRATION NUMBER

MANUFACTURER'S SERIAL NUMBER

FEDERAL STOCK NUMBER

CONTRACT NUMBER

DATE OF DELIVERY (MONTH AND YEAR)

GROSS WEIGHT

3.5.26 Part numbering and interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.5.27 Tiedowns. The vehicle shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport. The attachment poing shall be designed for use with attachment in the aircraft as described in in AFSC DH 1-11. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure. The tiedown points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook."

3.5.27.1 Clearance. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure on the truck. The tiedown points shall have a minimum clear opening of four inches to allow attachment of a chain tiedown with a grabhook.

3.6. Performance.

3.6.1 Pumping system operation. The pumping system shall exhibit no instability in pressure or flow rate under any operating condition. This system shall be capable of providing performance as specified below:

a. Pump from tank through the servicing hose at any flow rate of 0 GPM to 20 GPM. Once the pumping system has been started, flow shall be regulated on demand by control of the overwing nozzle.

b. Recirculate the contents of the tank at a flow rate of at least 15 GPM. Use the pump to mix the chemical required to disinfect the tank (for potable water usage), the propylene glycol antifreeze with demineralized water and alcohol with demineralized water.

c. Pump from the suction stub into the tank at a flow rate of at least 15 GMP using the suction hose attached to the suction stub.

d. Pressure servicing from an external source through the suction stub connection to the tank without operating the pump.

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3.6.1.1 Pump operation. The pump shall be of the self priming centrifugal type (see 6.3.4) or rotary type directly driven by a 12 volt DC electrical motor, a hydraulic motor, an air motor or power takeoff (PTO). This pump must be capable of transferring the tank contents at a rate of 15 GPM through the servicing hose when the nozzle is elevated 35 feet above the floor of the pumping compartment. The pumping system shall be designed so that the flow meter capacity cannot be exceeded in any mode of operation.

3.6.1.2 Power takeoff (PTO). At the option of the supplier, the pump can be driven by a PTO. The PTO shall be clutch operated (input gear continuously engaged to PTO drive gear) which enables the speed of the engine to be controlled preventing overspeeding of the pump. The pump and transmission are interlocked so that the pump shall not operate if the transmission is in any gear position other than neutral or park.

3.6.1.3 Closed discharge. The pump shall be capable of operating against a closed discharge for periods of 30 minutes without appreciable wear, damage or overheating. The closed discharge condition shall be with the pump motor operating a maximum allowable RPM and with all manual valves closed, except the tank shutoff valve (see 3.7.21.3). The water temperature in the pump shall not exceed 200 ^{o}F during the 30 minute period with an ambient temperature and initial tank water temperature of 75°F, plus or minus 5°F. This performance shall be attained with a maximum of 100 gallons of water in the tank.

3.6.1.4 Suction lift. The pump shall be capable of suction lift of 15 inches Hg to meet the test of 4.6.6.3.

3.6.1.5 Maximum flow. The pumping system shall be designed such that the maximum flow will not exceed the capacity of the flow meter.

3.6.1.6 Dry operation. The pump shall be capable of operation at maximum rated pump speed for 5 minutes without appreciable wear or damage after the unit has been completely drained.

3.6.2 Tank.

3.6.2.1 Static pressures. The tank shall withstand an internal static pressure of at least 3 PSIG without evidence of leaking.

3.6.2.2 Hydrostatic pressure. The piping system and all components shall withstand a hydrostatic pressure of 1 1/2 times the maximum pressure encountered during operation of the unit or a minimum of 100 PSI, whichever is greater,

3.6.2.3 Hydrodynamic loading. The tank shall be fillable at 200 gpm through a pipe in the tank manhole without damage.

3.6.3 Mobility. The truck shall be capable of operating while empty, and of transporting any rated payload up to 250 gallons of water over highways at a speed of 50 MPH and over unimproved roads at a speed of 40 MPH (see 6.3).

3.6.4 Suspension articulation. There shall be no interference, part failure nor permanent distortion of any of the truck when it is driven over a six inch high obstacle, either perpendicular to the direction the vehicle is traveling or at a 45° angle.

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3.6.5 Wheel loading. The difference in load between the left and right wheels of either axle shall not exceed three percent of the load on the axle for any cargo load condition. The difference between the curb weight and the fully loaded weight shall be "body payload." The distribution of the body payload between the front and rear axles (percent) shall be within the manufacturer's recommendation for the cab to axle dimension selected for this specification.

3.7 Details of components.

3.7.1 Chassis. The chassis shall be the manufacturer's standard, except as modified to comply with requirements herein. The chassis shall be 4 X 2 and shall conform to FED-STD-307, Type VII. The truck shall have a minimum payload rating of 3,500 pounds or equal to 110 percent of the weight of the fully loaded mounted equipment, whichever is greater.

3.7.2 Engine. The engine shall be a standard commercial diesel capable of providing the performance specified herein when operating on diesel fuel conforming to VV-F-800, JP-4 and JP-5 turbine fuel in accordance with MIL-T-5624.

3.7.3 Cab. The cab shall be of the low-cab-forward, cab-over-engine or cab-forward-of-engine or conventional type with a side by side seating arrangement to accomodate a crew of not less than two men wearing arctic clothing. The cab shall be designed to permit ease of maintenance, easy servicing of the engine, access to the sides and top of the engine while standing on the ground, and easy engine removal. The cab shall be of the all-steel (coupe) type with a comfortable, upholstered, full-width seat or individual seats, the driver's seat being fully adjustable. The color of the upholstering shall be compatible with the color of the exterior finish.

3.7.4 Cooling system. The chassis manufacturer's duty cooling system shall be provided.

3.7.5 Alternator. The alternator shall be of sufficient capacity to maintain an adequate supply of electrical power under all operating conditions for all 12 volt accessory components of the truck operating simultaneously. However, the alternator shall be not less than 80 amp capacity.

3.7.6 Battery. Battery capacity shall be sufficient to meet the low temperature starting and operating condition specified herein. The battery shall be 12 volts DC. The battery compartment shall be vented. The battery shall be of the maintenance free type.

3.7.7 Ammeter. An ammeter shall be provided and installed in the dash panel of the cab.

3.7.8 Hand throttle. A hand throttle shall be mounted in the dash panel of the cab for control of engine speed during all servicing operations. The throttle shall allow the operator to control the engine at any desired speed and in turn, control the output of the alternator to insure an adequate supply of electrical power. The throttle shall provide for sufficient movement and smooth operation. A turn to lock mechanism shall be provided to allow the throttle to be locked at any desired setting.

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3.7.9 Tires and wheels. Tires shall be standard commercial type tires. Standard commercial heavy duty type wheels shall meet the Tire and Rim Association requirements specified for this type vehicle. Unless specified (see 6.2), a spare tire, wheel, carrier and tools are not required.

3.7.10 Transmission. The transmission shall be automatic.

3.7.11 Parking brakes. The parking brakes shall hold the truck in any load condition in either direction on a slope of at least 20 percent.

3.7.12 Power steering. The truck shall be equipped with power steering.

3.7.13 Rear bumper. A full width, heavy duty rear bumper shall be mounted on the chassis frame. The bumper shall be designed to allow pushing of the fully loaded truck without damage.

3.7.14 Exhaust pipe and muffler. The truck shall be equipped with a spark arrestor muffler mounted underneath the chassis with the outlet pointing toward the curb. The muffler shall be so positioned and mounted that it cannot be easily damaged. The muffler shall be equipped with a cleanout plug.

3.7.15 Lighting and wiring system. Wiring lamps and all electrical equipment and components installed to the rear of the back of the cab shall conform to the following requirements.

3.7.15.1 Wiring. Wiring for lamps shall be no smaller than 12 gauge, stranded with thermoplastic insulation. Lamp pigtails, however, may be 16 gauge, but no longer than six inches. Wiring shall conform to SAE standards for low tension insulated cable, SAE J1128 and J1292. Wires shall be color coded or numbered to facilitate maintenance on the vehicle.

3.7.15.2 Wiring system installation. The wiring system shall be vaporproof. All wiring shall be enclosed in rigid, metal conduit firmly attached to the truck. All switches and circuit protection devices shall be of permanent vaporproof design or enclosed in vaporproof boxes. Wiring connections to these devices shall be through the metal conduit threaded to the component or vaproproof box. Wiring junctions shall be enclosed in vaporproof boxes with threaded connections to the metal conduit. The entire system and all components shall be designed to withstand a 3 PSI positive or negative pressure without leaking.

3.7.15.3 Protective devices. Overload circuit breakers shall be used in lieu of replaceable fuse type devices. All circuits shall be protected by circuit breakers. Protective devices shall be connected and identified in an organized manner to facilitate determination of malfunction.

3.7.15.4 Lamps. All lamps installed in the pumping compartment shall be of permanent vaporproof and waterproof design. The lamp body shall be nonferrous and shall incorporate a threaded fitting for making a vaporproof connection with the wiring conduit. The lens shall be heavy duty and capable of normal removal and installation without cracking or breaking and shall be designed to provide a vaporproof seal with the lamp body.

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3.7.16 Fenders. Fenders shall be provided over the rear wheels of the truck. The fenders shall be of heavy gauge steel and supported to withstand a static load of 250 pounds placed on top of the fenders. The outer edges of the fenders shall be rolled to a "J" shape or reinforced with rolled steel channel. Adequate clearance shall be maintained for easy installation of tire chains. Mudflaps shall be installed as needed to meet the requirements of SAE J682.

3.7.17 Winterization Kits.

3.7.17.1 Truck winterization. A standby type winterization kit designed to operate from an external 110 volt AC power shall be provided for the truck. The kit shall include at least immersion type heaters for the engine coolent, oil pan and transmission, as well as a battery box heater. The winterization kit shall be adequate for operation at -40°F including maintaining the engine in a standby condition capable of starting. The individual heater components shall incorporate automatic overheat controls as required. Glow plugs shall be furnished as standard equipment on the required diesel engine.

3.7.17.2 Potable water tank and pump compartment winterization. The cargo tank and pump compartment shall incorporate a standby type winterization kit designed to operate from an external 110 volt AC power. The heater in the cargo tank shall be controlled to limit the maximum water temperature to 60°F. The heater(s) in the pumping compartment shall be controlled to limit the maximum air temperature to +150°F. The individual heater components shall incorporate automatic overheat controls as required.

3.7.17.3 Receptacle, 110 volt. The 110 volt electrical components used for winterizing the truck shall lead to one receptacle securely mounted and clearly labeled on the outside of the cab. The 110 volt electrical components used for winterizing the tank and pumping compartment shall lead to a receptacle located adjacent to the aforementtioned receptacle and shall be clearly labeled. Each receptacle shall be of the male three prong twist lock type with a spring loaded dust cap included as part of its housing. Thr third wire shall be adequately grounded to the truck. There shall be no exposed conductor or terminals in the 110 volt systems which could be a hazard to personnel. At the option of the supplier, one receptacle can be supplied, provided a switch (circuit breaker) panel is installed so power distribution can be accomplished to each winterization circuit. Each switch shall be labeled with a permanent label.

3.7.17.4 Power on light. A red power on light shall be provided for each winterization system. These lights shall be located adjacent to each other on the dash panel and shall be clearly marked.

3.7.17.5 Circuit breaker. A manual reset circuit breaker shall be installed in each AC circuit inside the cab to afford overload protection.

3.7.17.6 Electrical cables. Three wire (ground) electrical cable/cables conforming to MIL-C-3432 shall be provided for each 110 volt receptacle. Each cable shall be 25 feet long. One end of each cable shall have a female plug which will mate with the male receptacle on the truck; the other end of the cable shall have a three conductor plug. The wiring shall be sized for the intended load and shall incorporate insulation impervious to water, alcohol, oil

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not in use.

3.7.18 Receptacle, 12 volt. An external receptacle conforming to MS 3506 shall be installed near the front right hand fender of the truck. A cover shall be provided and shall be held in the closed position by means of a spring. The receptacle shall be located in a relatively protected position and easily accessible for connection of heavy power cables. The receptacle shall be wired to provide for battery charging and starting of the truck engine by external power source. Conductors shall be sized to allow starting of the engine.

3.7.19 Tank. The tank shall be of the fully insulated single compartment type with a 250 gallon liquid capacity, plus at least a 10 percent expansion space. The tank shall be constructed of 300 series stainless steel and shall have sufficient thickness to provide the tank with complete structural integrity for the life of the truck. The tank shall be elliptical or modified rectangular cross section.

3.7.19.1 Drain. The tank bottom shall be sloped front to rear approximately one inch in three feet length to allow for complete drainage. A 1 1/2 inch drain valve shall be installed at the lowest point in the tank to permit complete drainage. The drain valve shall be protected against flying objects and to preclude freezing. The drain outlet shall be so designed that water cannot enter between the tank and the insulation cover. The valve control shall be extended for convenient operation of the valve without requiring the operator to go beneath the truck.

3.7.19.2 Baffles. Baffles shall not be provided in the tank unless required to provide longitudinal or transverse stability of the truck under any load condition. In the event surging is encountered on sharp cornering or in stopping or starting, baffles shall be provided. If provided, baffles shall have removable sections, or be designed to allow entrance into the tank and cleaning of all tank surfaces and corners.

3.7.19.3 Insulation. The tank shall be completely insulated (with the exception of the end facing the pumping compartment) with foam or a material that weighs not more than 1.5 PSF per inch of thickness. The material shall not deteriorate in water. The material shall be attached to the tank surface to prevent packing and settling. The insulation shall provide protection to preventa full tank of water, initially at 50°F plus 10°F, From freezing when exposed to an ambient temperature of -40°F for 12 hours.

3.7.19.4 Sheet metal covering. The insulating material shall be fully enclosed with an aluminum sheet metal covering which will form the external covering of the tank. The sheet metal covering shall prevent the entrance of water spilled around the manhole during loading and around the tank mounting due to road splash. The covering shall be reinforced as required to provide a rigid structure. The entire top surface of the tank and cover shall be capable of supporting two men weighing 200 pounds each standing at any point without permanent deformation of any portion of the tank and cover. The aluminum shall be insulated from the stainless steel tank to prevent electromotive corrosion.

3.7.19.5 Recirculation manifold. The pumping system recirculation line shall terminate in a stainless steel recirculation manifold installed in the tank. This manifold shall be designed to provide uniform mixing of the contents of

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the tank using the pump specified in 3.7.21.6.

3.7.19.6 Calibration marker. A fixed marker that is visible through the manhole shall be provided inside the tank to indicate when the tank has been filled with 250 gallons of water.

3.7.19.7 Manhole. The tank shall have a circular stainless steel manhole with a minimum diameter of 20 inches. The manhole shall have a 10 inch watertight hinged filler cover with necessary safety features to eliminate the possibility of a spark occuring during opening or closing of the cover. The manhole shall be equipped with a Buna N, or equivalent, gasket to insure a watertight seal between the collar and the cover. The entire top of the manhole shall be removable for entrance into the tank by removing a bolt on clamping ring.

3.7.19.8 Manhole well. The manhole assembly shall be installed in an insulated well. All insulation in the well and cover shall be completely enclosed and sealed to prevent entrance of water. The manhole and well shall be designed and constructed to preclude the entrance of water between the tank and sheet covering. The well shall have provisions for draining water which may accumulate during tank filling. The well shall not drain into the tank.

3.7.19.8.1 Cover. A hinged, insulated cover shall be provided to allow for convenient access to the manhole. All insulation in the cover shall be completely enclosed and sealed to prevent entrance of moisture. An automatic engaging device shall be provided to hold the cover in the open position. This hold open device shall be manually disengaged. The cover, hold open device, and mounting hinges shall be adequately constructed to prevent damage under high winds. Two quick disconnect latches shall be utilized to hold the cover in the closed position. The latches and mounting hinges shall be replaceable with common hand tools.

3.7.19.9 Vent. A vent shall be provided to allow water to be pumped into and out of the tank at 40 GPM without exceeding 1 PSI pressure or vacuum. The vent may be an open tube protected to preclude the entrance of atmospheric contaminants. The vent shall be so designed that water will not splash out under any operating condition or become plugged due to ice, snow or other contamination.

3.7.19.10 Inspection platform. A platform shall be provided on top of the tank to permit convenient operation of the well cover and manhole cover. The platform shall be at least 15 inches by 18 inches and shall incorporate an expanded metal, or equivalent, permanent nonskid surface. The platform shall support a load of 350 pounds without permanent deformation.

3.7.19.11 Access provisions. A ladder section, or steps and handrails, or grag handles shall be provided on the rear of the truck to permit safe and convenient access to the platform from the ground under all operating conditions. The ladder shall be constructed of tubing at least one inch in diameter and capable of supporting a 250 pound operator with no noticeable deflection. The ladder shall be at least 15 inches wide.

3.7.20 Pumping compartment. The pumping system shall be housed in a completely enclosed compartment located either behind or in front of the cargo tank. In either location, the pumping controls, servicing hose and nozzle storage box shall be accessible for normal operation. One side of the compart-

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ment shall be formed by one end of the cargo tank to facilitate warming of the pumping compartment. The junction between the tank and compartment shall incorporate an effective seal and result in a minimum of heat loss.

3.7.20.1 Pumping compartment construction. The pumping compartment shall be galvanized steel of double construction and insulated, including the doors. All joints shall be tight, and the insulating shall be fully enclosed. The insulation shall be at least one inch thick fiberglass, foam, or equal, and be completely enclosed in sheet metal. All panels shall be integrally constructed with reinforcement and permanently welded. The insulating material shall be cemented in place to preclude the possibility of packing and settling. The roof shall be removable. All doors and hinged sections shall have full length piano type stainless steel hinges. The compartment assembly shall incorporate neessary structural integrity to prevent bending, warpage or other damage due to vibration and forces encountered during normal operation. The mounting hinges shall be replaceable with common hand tools. Sharp edges and corners which could be a hazard to operating personnel are not allowed. The pumping compartment shall be designed and constructed to preclude freezing of water-bearing plumbing and components when exposed to an ambient temperature of -40°F for one hour with an initial compartment air temperature of 55°F.

3.7.20.2 Roof. The roof shall be removable as an assembly by one man using common hand tools within 10 minutes. The roof shall support two men weighing 200 pounds each standing at any point without permanent deformation. The roof panel shall incorporate lifting handles to facilitate removal and installation.

3.7.20.3 Access doors. Three doors shall be provided at the rear or driver's side of the compartment depending upon compartment location so that the pumping system components and controls are accessible for viewing, adjustment and removal. One vertically hinged door shall provide access to the control panel and nozzle storage box. One vertically hinged door shall be provided adjacent to the control panel door to provide for inspection and maintenance of a major portion of the pumping system. This door shall remain closed during low temperature operation, heating system operation and servicing. The third door shall be located below one of the other doors and hinged to open downward. The third door shall be minimum size commensurate with safe and convenient extension and rewinding of the hose assembly, including the nozzle with the spout attached.

3.7.20.4 Vertically hinged door(s). If the pumping compartment is located between the cargo tank and the cab, a vertically hinged door shall be provided on the curb side of the compartment to provide for clear access to at least 75 percent of the area of the side of the compartment. However, if the pumping compartment is placed behind the cargo tank, vertically hinged doors shall be provided on both the driver's side and curb side of the compartment to provide for clear access to at least 75 percent of the area of each side of the compartment.

3.7.20.5 Hold open devices. All doors shall be provided with automatic engaging devices to hold the doors in the hold open position. The doors and hold open devices shall be adequately constructed to prevent damage under high winds. The hold open devices may be manually disengaged.

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3.7.20.6 Door latches. Compartment door latches shall be rigidly constructed and mounted to prevent bending, distortion or other type failure during long periods of operation. The latches shall be of the quarter turn release type with a positive locking means which will not allow the latches to release inadvertently due to vibration or other forces. The door latches shall be so mounted as to be replaceable with common hand tools.

3.7.20.7 Locking pins. Two locking pins per latch shall be provided, one at the top and one at the bottom of the door. Locking pins shall engage the panel structure rather than the opposite door. The latches and locking pins shall be designed to allow operation by personnel wearing arctic mittens.

3.7.20.8 Floor. A drain shall be provided in the floor to permit complete drainage of the compartment. The drain shall be easily accessible and adequately sized to allow drainage in the minimum amount of time. A galvanized sheet metal sleeve shall be provided for each hole cut in the floor. The sleeve shall be fastened to the top and bottom galvanized sheets covering the floor in such a manner as to permanently prevent entry of moisture.

3.7.20.9 Ventilating provisions. The pumping compartment shall be designed to provide for natural air circulation to preclude a high humidity condition which could exist if the compartment were sealed for an extended period of time. Provisions shall be incorporated for covering the vent(s) for low temperature operation and compartment heating. These coverings shall be permanently attached to the compartment or shall have storage provisions within the compartment. There shall be no ventilating openings in the roof or floor of the pumping compartment.

3.7.21 Pumping system. The pumping system shall include the tank to pump suction line, pump, tank return line, filter, hose reel, servicing hose, nozzle, valves, control panel, associated plumbing and minor components. The pumping system shall be capable of operation totaling 140 hours without degrading effects.

3.7.21.1 Tank to pump suction line. The tank to pump suction line shall be designed to provide adequate flow and smooth operation of the pump under all performance conditions required in this specification. The tank to pump suction line shall incorporate a Y type line strainer, tank shutoff valve, suction stub and, if required by the design of the tank and pumping compartment mountings, a flexible coupling. The tank to pump suction line shall terminate in the bottom of the tank, except where electric emmersion heaters are used. The tank to pump suction line shall terminate in the end of the tank to allow an adequate quantity of water to cover the heater elements to preclude element burnout.

3.7.21.2 Line strainer. The Y type line strainer shall be a rigidly constructed 20 mesh screen adequately sized to allow a 20 GPM flow at minimum pressure drop. The screen shall be easily removable for cleaning and inspection. The strainer shall be accessible through one of the pumping compartment doors. The strainer shall be so mounted that a container can be located beneath the strainer to prevent spillage into the compartment. the pumping system shall be so designed that water will not drain therefrom when the screen is removed.

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3.7.21.3 Tank shutoff valve. The tank shutoff valve shall be of the quarter turn, full on to full off type. The valve shall be located between the tank and the line strainer.

3.7.21.4 Suction stub. The suction stub shall be a nominal 1 1/2 inch size with a manually operated shutoff valve. The stub shall be provided with a 1 1/2 inch female cam locking type stainless steel coupling and dust plug in accordance with MIL-C-27487. The security cable for the dust plug shall be corrosion resistant wire rope. The suction stub shall be located upstream of the line strainer.

3.7.21.5 Flexible coupling. If required to provide relative motion between the pumping system and the tank, flexible couplings shall be located downstream of the tank cutoff valve. The couplings shall withstand an internal pressure up to three times the working pressure or 50 PSI, whichever is greater, and a vacuum of 0 inches Hg to 15 inches Hg without leakage or other failure.

3.7.21.6 Pump. The pump shall be of the self priming centrifugal type or a rotary type directly driven by a motor or a power takeoff unit. The rotary type pump shall include a bypass type relief valve to prevent discharge pressure above the system design pressure. This discharge nozzle pressure shall be limited to a maximum of 30 PSI.

3.7.21.7 Motor. If an electric motor is used, it shall conform to the explosion proof requirements of NFPA Pamphlet 70 as specified for Class 1, Division 2, Group D hazardous locations. The motor shall utilize the 12 volt DC truck electrical system and shall be sized to comply with the requirements specified herein.

3.7.21.7.1 Overspeed protection. The motor shall be protected from overspeed in the event the water system is pumped dry.

3.7.21.8 Pump seal failure. The pump shall be so constructed and positioned that, in the event of pump seal failure, water will not enter the pump motor.

3.7.21.9 Pump drain. The pump shall be so designed that it can be completely drained. The drain plug shall be replaced with a drain line and valve capable of completely draining the pump without the use of hand tools. The drain line shall extend beneath the floor of the pumping compartment.

3.7.21.10 Tank return line. The pumping system shall incorporate a tank return line between the pump outlet and the tank to permit recirculation of the contents.

3.7.21.11 Return line shutoff valve. The tank return line shall incorporate a shutoff valve located between the tank and the flexible coupling of the first detachable joint.

3.7.21.12 Filter. A filter shall be installed in the pumping system downstream of the pump. All fluids dispensed from the servicing hose shall have passed through the filter. The filter shall incorporate one element in accordance with MIL-F-83428. The filtering screen shall be stainless steel.

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3.7.21.13 Air eliminator. The filter shall incorporate an air eliminator to allow automatic purging of air from the pumping system. The air eliminator shall be vented into the cargo tank.

3.7.21.14 Filter drain. The filter shall be provided with a hand operated drain valve capable of draining the complete contents of the filter. The drain line shall extend beneath the floor of the pumping compartment. A galvanized sheet metal sleeve of at least 20 gauge material shall be provided for each hole cut in the floor. The sleeve shall be fastened to the top and bottom galvanized sheets covering the floor in such a manner as to permanently prevent entry of mositure into the floor.

3.7.21.15 Meter. A 20 GPM positive displacement meter shall be installed downstream of the filter in the pumping compartment. The meter shall be of a commercial design that has been manufactued in production qualities, and the material and environmental requirements specified herein shall be compatible with the liquids to be serviced by this truck. The meter shall be accurate within plus or minus one percent at all flow rates between 5 GPM and 25 GPM. The maximum pressure drop between the meter inlet and outlet shall not exceed 3 PSI when operating at rated capacity at 70°F.

3.7.21.15.1 Register. The meter register shall contain a set of at least three direct reading digits. Each digit shall have a minimum height of 1/4 inch. Each digit shall be resetable to zero and capable of recording a flow of at least 999 gallons of liquid. The register shall be clearly visible and within easy reach of an operator standing on the ground at the control panel.

3.7.21.16 Servicing hose reel. An open type servicing hose reel shall be installed in the pumping compartment. The hose reel shall be of commercial design that has been produced and supplied as a part of fluid servicing system. The hose reel shall conform to the material requirements specified herein and shall be suitable for servicing potable water, demineralized water or water/alcohol mixtures at a flow rate of at least 20 GPM. The hose reel hose may be of the single wrap or multiwrap type so designed that the servicing hose may be readily removed without dismantling parts of the reel. The inlet connection to the hose reel shall be of the groove type conforming to MIL-C-10387.

3.7.21.16.1 Servicing hose. Fifty feet of one inch ID noncollapsible Class 2, Grade 3 servicing hose conforming to ZZ-H-601 shall be provided on the reel. The hose shall have stainless steel male quick disconnect couplings conforming to MIL-C-27487 on each end. The coupling on the servicing end of the hose shall be provided with a dust cap. A security cable for the dust cap shall be 1/16 inch diameter wire rope in accordance with MIL-W-83420 except with a clear nylon jacket.

3.7.21.16.2 Spring motor. A durable spring motor shall provide for automatically rewinding the hose onto the reel. A safety decluctching device shall be provided to protect the spring motor from damage or fatigue in the event of severe backlash or reverse rotation. The reel shall allow for the extension of the complete length of hose upon demand, and a latching mechanism shall be provided that will prevent the hose from rewinding onto the reel. Upon release of the latching machanism by further extension of not more then one foot of hose, the hose filled with water shall completely rewind onto the reel. The hose

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reel shall require an unwinding force of no more than 15 pounds. This force shall be measured as a direct pull on the hose when one half of the hose length is extended from the reel. The hose reel shall be capable of performing 1,500 hose extension cycles (see 6.3) without failure to any component. Fracture, cracking, binding or sereve wear of any part or component of the reel shall be considered a failure.

3.7.21.16.3 Hose rollers. Four way stainless steel rollers shall be provided to prevent the hose from rubbing the frame or sharp corners when beig unreeled or rewound.

3.7.21.16.4 Hose stop. A rubber hose stop shall be provided to protect the nozzle from damage when the hose is rewound onto the reel.

3.7.21.17 Nozzle and servicing adapters.

3.7.21.7.1 Nozzle. A one inch nominal size water servicing nozzle shall be provided. The body of the nozzle, valve, main spring and other operating parts shall be stainless steel. The nozzle squeeze control valve shall be capable of throttling the liquid flow from 0 GPM to 20 GPM without adjustment of the pumping system. The inlet connection of the nozzle shall be a one inch female quick disconnect coupling, and the outlet connection shall be a one inch male quick disconnect coupling, fabricated of stainless steel in accordance with MIL-C-27487. The nozzle. shall withstand a static pressure of 100 PSI or 1 1/2 times the maximum total pressure encountered during operation of the unit, whichever is greater.

3.7.21.17.2 Spout. The nozzle shall be provided with a one inch removable curved stainless steel outlet spout. The spout shall be a minimum length of six inches. A removable strainer shall be incorporated into the spout. The conical shaped screen shall be fabricated of 300 series stainless steel, and the screen size shall be 100 mesh. The screen shall be attached to a reinforcing ring installed in the inlet end of the spout ans secured to prevent the liquid from bypassing around the strainer. The spout shall be provided with a one inch stainless steel female quick disconnect coupling in accordance with MIL-C-27487 for attachment to the nozzle.

3.7.21.17.3 Protective cap. The nozzle shall be provided with a protective cap that will prevent the entrance of dirt or othe foreign matter when not in use.

3.7.21.17.4 Pressure servicing adapters. The following pressure servicing adapters shall be provided with each truck:

a. One inch size. A stainless steel adapter with a one inch stainless steel female quick disconnect coupling in accordance with MIL-C-27487 shall be provided for connecting the nozzle to the one inch inlet fitting, Roylyn Part Number 1002-J-16-C, quick disconnect coupling, on the potable water tank of the type A/B-375-1 buffet lavatory unit.

b. Three quarter inch size. A stainless steel adapter with a one inch stainless steel female quick disconnect coupling in accordance with MIL-C-27487 shall be provided for connecting the nozzle to the 3/4 inch inlet fittings,

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Roylyn Part Number 1002-12 or 1309-12, quick disconnect coupling, on the potable water tanks of the Types C-141 and C-5 Aircraft.

3.7.21.17.5 International. A stainless steel adapter with a one inch stainlesssteel female quick disconnect coupling in accordance with MIL-C-27487 shall be provided for connecting the nozzle to the aircraft drinking water replinshment connection as shown on MS 21391 (see 6.5).

3.7.21.18 Storage box. A stainless steel storage box shall be provided in the heated pumping compartment. The storage box shall house a container for storing the nozzle specified in 3.7.21.17.1, the spout specified in 3.7.21.17.2 and the adapters specified under 3.7.21.17.4 in a chlorine or othe type disinfectant solution. The storage box shall be easily removed and held in place by two or more latches or clamps. The storage box shall be equipped with handles. The nozzle, spout and adapters shall be easily withdrawn from or added to the storage box while the box is held in its normal position in the

3.7.21.18.1 Cover. The storage box cover shall be hinged and vapor tight. A means shall be provided for holding the cover in the open position and two or more latches provided for holding the cover in the closed position. It shall be possible to move the cover from its closed position to its full open position while the storage box is held firmly in its normal position in the pumping compartment.

3.7.21.18.2 Container. A stainless steel tray type container shall be provided so the operating personnel can remove the stored compartments without inserting their hand(s) in the solution. The storage box and container shall be designed to prevent excess sloshing of the solution.

3.7.21.19 System operating valves. Valves shall be provided as necessary for safe efficient operation of the pumping system and to meet the performance and functional requirements specified herein. The pumping system shall incorporate a tank shutoff valve, suction stub valve and return line shutoff valve as a minimum. The tank shutoff valve shall be of the quarter turn open and close type, or of the quick opening gate type. All manual valves shall be located within one foot of a door within easy reach and in clear view of the operator standing on the ground. All valves shall be easily operated when the pumping system is under normal operating pressure.

3.7.21.20 Drain valves. Manually operated drain valves shall be provided as required to insure complete drainage of all portions of the pumping system and all components except the hose. Water in the hose may be drained from the nozzle end. The drain valves shall be easily accessible and adequately sized to allow drainage in the minimum amount of time. The drain lines shall extend beneath the floor of the pumping compartment.

3.7.21.21 Suction hose. Five feet of 1 1/2 inch ID suction hose conforming to ZZ-H-561, Grade A, shall be provided. The suction hose shall have a male quick disconnect coupling on one end and a female quick disconnect coupling on the other end, both of which shall conform to MIL-C-27487, except the material shall be of stainless steel. The suction hose also shall be provided with a stainless steel dust cap and plug in accordance with MIL-C-27487. The security cables for the dust cap and plug shall be 1/16 inch diameter corrosion resis-

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tant wire ropes.

3.7.21.22 Suction tube assembly. A suction tube assembly, complete with an airlock nozzle shall be provided. The airlock nozzle shall shut off flow when the storage drum is empty. The suction tube shall be connected to the airlock nozzle and designed for insertion into a storage drum. A screen shall be provided at the bottom of the tube to prevent solid contaminant from reaching the airlock nozzle. The suction tube assembly shall be provided with a stainless steel male cam locking quick disconnect coupling, half with a dust cap in accordance with MIL-C-27487. The security cable for the dust cap shall be a 1/16 inch diameter corrosion resistant wire rope.

3.7.21.23 Stowage compartment. A dust tight stowage compartment shall be provided to accomodate the complete suction hose and suction tube assembly. The compartment shall be fitted with a door that will open wide enough to provide access and easy removal of the components stowed herein. The bottom of the compartment shall be sloped down toward the opening to allow any water remaining in the hose or suction tube to drain away.

3.7.21.24 Control panel. A control panel shall be located within the compartment within easy reach of, and clearly visible to, an operator standing on the ground. The control panel shall be accessible through the compartment door. The rear of the control panel shall be easily accessible for maintenance. The control panel shall contain a pumping compartment light switch, engine emergency switch, pump electric motor switch and indicating light, tank gauge and water temperature indicator.

3.7.21.25 Pressure guage connection. A pressure gauge connection shall be procided in the pump outlet line.

3.7.21.26 Engine emergency switch. The engine emergency manual cutoff switch shall provide for immediate and complete shutdown or the truck engine from any operating condition without the operation of any other controls. The switch shall be circled with red paint one to two inches in diameter or otherwise conspicuously marked.

3.7.21.27 Pump electric motor switch. An on/off switch shall be provided to control the electric motor which drives the dispensing pump. A yellow indicating light shall be located adjacent to the pump on/off switch and shall be illuminated when the switch is in the on position.

3.7.21.28 Tank gauge. A gauge that indicates the number of gallons of water in the tank shall be provided on the control panel. The gauge shall have an accuracy of plus or minus five percent.

3.7.21.29 Water temperature indicator. A tank water temperature indicator shall be installed on the control panel. The indicator shall be scaled from 30°F to 120°F in increments of 5°F.

3.7.21.30 Piping. The piping layout shall be designed to facilitate maintenance on the pumping system and removal of components with minimum disassembly of piping. Grove type couplings conforming to MIL-C-10387 shall be used to the maximum practical extent. Piping sections shall be easily removable for main-

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tenance and repair. Piping connections to major and minor components shall be of the groove type to the maximum practical extent. The piping shall be designed to facilitate draining. Pipe sections that are subject to excessive movement during operation shall be firmly mounted or braced. Piping wall thickness shall be minimum of Schedule 40.

3.7.21.31 Tubing and conduit. Static pressure and guage sensing tubing lines and electrical conduit shall be installed in groups wherever possible and mounted on blocks which provide for separation of the lines and mounting of multiple lines in one place on any flat surface.

3.7.21.32 Static discharge grounding reels. One RMU/25E reel conforming to MIL-R-83232/2 and one RMU.26E reel conforming to MIL-R-83232/3 shall provided. The reels shall be installed side by side on the driver's side of the truck in a readily accessible location. The location of the reels shall permit easy withdrawal and recovery of cables. The reels shall be bolted to the truck and the resistance between the reels and the chassis frame and tank shall not exceed 0.5 OHM.

3.7.21.33 Pump compartment heating system. An electrical or electrical/hot water space heater shall be provided to heat and maintain the pumping compartment in a heated condition. The heating system shall be capable of operating while the truck is being driven and during the servicing operation. Operation of the heating system for extended periods shall not cause thermal degradation of adjacent equipment or any portion of the truck.

3..7.21.33.1 Heating system performance. The heating system shall maintain a compartment temperature of at least 55°F with an ambient temperature of -40°F.

3.7.21.33.2 Heater operation. The heater shall operate automatically without attention for long periods of time. The heater and controls shall be designed to be fail safe to the maximum practical extent.

3.7.21.33.3 Heater protection. Protective devices (i.e. thermostats, overload controls, etc) shall be installed to insure that excessive temperatures and pressures will not be experienced in any component.

3.7.21.33.4 Personnel protection. Protective devices (i.e. expanded metal guards) shall be installed around the heater elements to preclude injury to the operating personnel.

3.7.21.33.5 Heater electical controls. The heater electrical controls shall operate on 12 volt DC power from the chassis electrical system. The electrical switch required for operating the heater shall be mounted on the dash panel of the truck cab. A green indicating light shall be located adjacent to this switch and shall be illuminated when the switch is in the on position.

3.7.21.33.6 Heating system electrical installation. The entire heating electrical system shall be separate from the truck electrical system so that the heating system can be removed without disturbing the truck electrical system.

3.7.22 Lighting.

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3.7.22.1 Pumping compartment lights. At least two pumping compartment lights shall be furnished. The lights shall illuminate the entire interior of the compartment and allow observation of all controls and instruments. The lights shall be vaporproof and shall incorporate threaded conduit connection.

3.7.22.2 Night servicing lights. The truck shall be provided with at least two fully adjustable lights to facilitate servicing in the darkness. Both lights shall have switches on the adjusting handle. The night servicing lights shall not operate unless the truck driving lights or parking lights are turned on.

3.7.22.3 Forward left corner of cab. One light shall be standard automotive spotlight mounted in the forward left corner of the driver's side of the pumping compartment of the cab. The light shall incorporate ANSI Number 4435 lamp. The light shall be so installed that it can be adjusted by the driver, but does not interfere with other normal driving functions, or entering, or exiting the cab.

3.7.22.4 Top left rear corner of truck. A light shall be located near the top left rear corner of the truck. The light shall be at least six feet from the ground. The light shall be adjustable by an operator standing on the ground. The light shall be so mounted that it will illuminate the rear or entire left side of the truck. The light shall incorporate an ANSI Number 4412 lamp. The thermoplastic insulated wiring from the light assembly to the rigid vaporproof conduit shall be enclosed in a protective flexible metal conduit.

3.7.23 Fire extinguisher. One 20 pound Halon 1211 fire extinguisher, which is Underwriter's Laboratory (UL) approved, shall be mounted vertically in a bracket on the driver's side of the truck. Protection shall be provided from the splashing of mud and water. The fire extinquisher shall be readily accessible to personnel standing on the ground and shall not be mounted behind any pumping compartment door when the door is open.

3.7.24 Acoustic noise levels. The maximum A-weighted sound level produced by the vehicle during pumping operations shall not exceed 85dBA when measured as specified in 4.6.15.

3.7.25 Dimensions. The overall dimensions specified below shall be further reduced as required based upon the exact configuration of the truck to be furnished and the air transporability requirements specified herein.

3.7.25.1 Width. The overall width of the truck shall not exceed 96 inches.

3.7.25.2 Height. The overall height of the truck shall not exceed 85 inches in the empty condition.

3.7.25.3 Length. The overall length of the truck shall not exceed 18 feet.

3.7.26 Diagrams. Schematic diagrams of the piping and electrical system shall be provided. Each valve, switch, etc, on the diagram shall be properly identified to correspond to the markings on like parts in the truck.

3.7.27 Marking of tiedown points. The location and capacity of each tiedown

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point specified in 3.5.27 shall be clearly marked.

3.7.28 Dimensional tolerances. Dimensions and tolerances nct specified shall be as close as consistent with the shop practices. Where dimensions and tolerances may affect the interchangability, operation or performance of the truck, they shall beheld or limited accordingly.

3.7.29 Piping. The installation of piping, tubing, writing, etc, shall be so accomplished as to prevent rubbings and wear due to viberation. The tubing and plumbing shall be assembled to permit adequate flexibility under torsional forces induced during operation. Failure in piping shall not occur as a result or road operation or torsional forces.

3.8 Workmanship. The vehicle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges, accuracy of dimensions, surface finish and radii of fillets, throughness of soldering, welding, brazing, painting, wiring and riveting, marking of parts and assemblies, alignment of parts and tightness of assembly fasteners, etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the Contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products' or supplies' submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the vehicle shall be classified as follows:

a. First article testing (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

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4.3.2 Lubricants. Lubricants used in the truck during the tests specified under 4.4.4 shall be as recommended by the truch chassis manufacturer and the contractor for the mounted equipment and shall conform to existing Government specifications for the type services.

4.3.3 Test fluid. Unless otherwise specified, the test fluid shall be tap water. The fluid used in the test specified in 4.6.6.4 shall be demineralized water with corrosion preventative conforming to MIL-C-4339. This corrosion preventative shall be mixed with the demineralized water at a rate of 2/3 of one percent of the total water volume.

CAUTION

It is imperative that the oil and water be throughly mixed before adding alcohol and that all residue of a previous batch be removed from the mixing unit before preparing a new mixture, because the oil is soluble in water, but will not dissolve in alcohol or a premixed water/alcohol mixture.

Type I detergent conforming to MIL-D-16791 shall also be mixed with the above demineralized water as a wetting agent in the amount of one quart per 1,000 gallons of the total mixture to be prepared.

4.3.4 Data. During the testing specified herein, at least the following data shall be recorded at intervals not greater than 30 minutes, except where additional data and shorter time periods are a test requirement:

a. Date.

b. Time started.

c. Time Finished.

d. Ambient temperature.

e. Water volume in cargo tank (gallons).

f. Water tank temperature.

g. Pressure at pump inlet (in Hg).

h. Pump discharge pressure.

i. Engine speed (RPM).

j. Engine oil temperature.

k. Engine coolant temperature.

l. Nozzle pressure.

m. Flow rate.

4.3.5 Preparation for test. The vehicle equipped with all specified auxiliary

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equipment shall be fully serviced, and all equipment shall be prepared for immediate operation.

4.3.6 Test rejection criteria. Throughout all test specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel, coolant, lubrication or hydraulic fluid, under any operating condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking and while traversing rough terrain.

h. Conditions which present a safety hazard to personnel during operation, service or maintenance.

i. Overheating of the engine, transmission or any other vehicle components.

4.4 First article testing.

4.4.1 First article test sample. The first article test sample(s) may be a preproduction item(s) or production item(s) which meets the requirements of this specification. The sample(s) shall be identical to the items procured under the terms of the contract. Approval of the first article sample shall not constitute a waiver by the Government of its rights under the provisions of the contract. Two first article samples shall be produced, when specified (see 6.2), and tested as follows. The first article vehicle(s) shall be retained as an example.

4.4.2 First article test. The first article tests shall consist of all the tests described in 4.6.1 through 4.6.22.3.

4.4.3 Test report. Upon completion of the first article test, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report.

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The following shall be incorporated as a part of the test report:

a. All failures, maintenance performed and other events recorded shall be identified by accumulated operating time, miles, cycles or position in the test procedure as appropriate. The conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made, or proposed to be made and the predicted effeciveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808.

g. List of methods to be used to protect against electrolytic corrosion of dissimilar metals in intimate contact in accordance with MIL-STD-889.

h. Test reports or certification and date of first article approval for all components covered by detail specifictions requiring first article testing.

i. Plan for testing each production unit.

4.4.4 Preproduction tests. The preproduction tests shall consist of all the tests described in 4.6. If failure occurs as a result of any one test, the test program shall be temporarily stopped and the failure reported to the procuring activity. Inability to satisfactorily complete any of the tests specified under 4.6 and to meet any of the associated requirements specified in Section 3 shall constitute a failure. A failure shall also be defined according to 6.3 except failure action, including the amount and sequence of further testing, will be as directed by the procuring activity.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the following:

a. Individual tests (see 4.5.1).

b. Periodic sampling (see 4.5.2).

c. Operational capability check (see 4.5.3).

4.5.1 Individual tests. Each truck shall be subjected to the following tests

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as described under 4.6:

a. Examination of product (see 4.6.1).

b. Mechanical and electrical operation (see 4.6.2).

c. Static pressure (see 4.6.3).

d. Pumping system hydrostatic test (see 4.6.5).

4.5.2 Periodic sampling. One truck shall be selected at random from the first 10 manufactured on a production contract. Thereafter, one truck shall be selected at random from every 50 or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Pumping system flow tests (see 4.6.6).

b. Road handling test (see 4.6.20.1).

4.5.3 Operational capability check. Sample trucks shall be selected at random. Sample size shall be in accordance with MIL-STD-781. Servicing cycles (subparagraphs a through f) shall be conducted on each selected truck in the following sequence:

a. Truck tank shall be filled to capacity with potable water at a temperature of 60°F, plus or minus 5°F.

b. Truck engine shall be started.

c. Vehicle shall be driven a minimum distance of 1/2 mile over improved (paved highway) or unimproved (graded gravel) roads to an empty storage tank and stopped.

d. Servicing hose shall be extended 45 feet and 135 gallons of water at 20 GPM shall be pumped into the empty storage tank.

e. Vehicle shall be driven back a minimum distance of 1/2 mile for refilling.

f. Truck engine shall them be stopped, and the truck shall be refilled to capacity.

4.5.4 Random sample testing. Any production vehicle may be selected by the procuring activity at any time to be subjected to the tests described in 4.6. Upon completion of the random sample testing, the contractor shall prepare a test report in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets of the report.

4.5.4.1 Rejection and retest. When a vehicle fails to meet the requirements of this specification, vehicles to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined, corrective action is taken by the contractor and the component that failed is successfully retested to assure compliance with this specification. Tests not involving the

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failed component may be continued pending the investigation of a test failure.

4.6 Test methods.

4.6.1 Examination of product. The truck shall be inspected to determine compliance with the requirements in this specification. Particular attention shall be given to compliance with design requirements on preproduction trucks. Particular attention shall be given to materials, workmanship, dimensions, protective finishes and their application, welding, riveting and markings. A checklist, containing a reference to all applicable requirements of Section 3 of this specification, shall be completed as evidence of this test.

4.6.2 Mechanical and electrical operation. A thorough check shall be made of the truck to insure the proper functioning of all mechanical and electrical equipment, accessibility of all controls and levers, interchangeability of removable parts, installation of equipment without fitting and to effect every degree of safety and performance. A checklist, containing a reference to all applicable requirements of Section 3 of this specification, shall be completed as evidence of this test.

4.6.3 Static pressure. The truck shall be subjected to an internal static air pressure of 3 PSIG for not less than 30 minutes. A soap solution shall be sprayed on the exterior of the tank to facilitate detection of leaks that may exist. Evidence of leakage shall be cause for rejection.

4.6.4 Vent operation. The vent operation test shall be conducted after the tank has successfully accomplished the static pressure test of 4.6.3. All tank openings, except the vent and manometer pressure tap, shall be effectively sealed. Tank pressures shall be recorded at 15 second intervals, beginning when flow is started into and out of the tank and continuing until the respective plus or minus pressures in the tank stabilize. Water shall be pumped into and out of the tank at 40 GPM. The respective maximum positive and negative pressures shall be reported and shall not exceed 1 PSI.

4.6.5 Pumping system hydrostatic test. The flow system and component of the pumping system on the pressure side of the pump shall be subjected to a 30 minute hydrostatic test of 1 1/2 times the maximum total pressure encountered during operation of the unit or 100 PSI, whichever is greater. There shall be no evidence of leakage.

4.6.6 Pumping system flow. All pumping system flow tests shall be accomplished without performing adjustment to the pumping system other than manipulation of controls by the operator in accordance with the operating instructions.

4.6.6.1 Test fluid. The test fluid shall be pumped through the complete pumping system of the truck into a prover tank at flow rates from 5 GPM to 25 GPM in 5 GPM increments. The maximum error at any flow rate over this range shall not exceed plus or minus one percent.

4.6.6.1.1 Maximum flow test. The test fluid shall be pumped through the complete pumping system of the truck into a prover tank to determine the maximum flow rate attainable by manipulation of the operator controls, including the

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engine hand throttle. The maximum attainable flow rate shall be recorded and shall not exceed the flow rating of the flow meter.

4.6.6.2 Pressure differential test meter. The test fluid shall be pumped through the meter at flow rates from 5 GPM to 25 GPM in increments of 5 GPM. The pressure drop between the inlet and outlet of the meter shall be measured after each flow rate has been stabilized for two minutes. The maximum pressure drop between the inlet and outlet of the meter during this test shall not exceed 3 PSI. The actual pressure measurements shall be recorded at each required flow rated from 5 GPM to 25 GPM.

4.6.6.3 Flow through overwing nozzle. The nozzle attached to the servicing hose shall be elevated to 35 feet above the floor of the pumping compartment. Suitable piping shall be arranged to recirculate the water into the manhole of the cargo tank. A flow rate of at least 15 GPM through the hose shall be demonstrated. By controlling only the overwing, nozzle, flow rates of 0 GPM, 5 GPM, 10 GMP and 15 GPM shall be demonstrated without adjustment or mainpulation of the pumping system controls. Each flow condition shall be maintained for two minutes. There shall be no instability in the pumping system operation. The pumping system shall demonstrate rapid flow recovery when the nozzle valve is opened after complete shutoff.

4.6.6.4 Dedrummin. The capability of on loading ethyl alcohol conforming to O-E-760 Type III at a minimum flow rate of 15 GPM into the truck tank and the capability of the airlock nozzle to automatically shutoff when the drum is empty shall be demonstrated. The tank shall be initially filled with 190 gallons of demineralized water as specified in 4.3.2 prior to adding the alcohol. A total of 60 gallons of alcohol shall be withdrawn from two drums through the suction tube assembly, suction hose and suction stub. The top of the storage drum shall be at least three feet below the centerline of the pump impeller shaft. Following the transfer of the alcohol into the tank, the pump shall be immediately shut down. The recirculation test (see 4.6.6.5) shall then be conducted.

4.6.6.5 Recirculation. Following the completion of the dedrumming test (see 4.6.6.4) the recirculation test shall be initiated by starting the pump and demonstrating that the pumping system will recirculate the contents from the cargo at a flow rate of at least 15 GPM without the use of the servicing hose. After mixing the water and alcohol for 30 minutes, two fluid samples shall be extracted from the front and rear sections of the tank. Specific gravity readings of the samples shall be equal.

4.6.7 Closed discharge. The pump shall be operated against a closed discharge as specified in 3.6.1.3. The water temperature in the pump shall not exceed 200°F, and the pump shall not have any appreciable wear or damage.

4.6.8 Dry operating. The pump and pumping system shall be completely drained. Without introducing liquid into the pumping system, the pump shall be operated for five minutes. There shall be no appreciable wear, damage, overheating or other type failure as a result of this test.

4.6.9 Pumping system endurance. With the truck engine operating, the pumping system shall be operated at rated flow for at least 100 hours. The system

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shall be operated continuously for periods of no less than six hours throughout the 100 hour endurance test. Elapsed time between continuous operation shall not be considered in accumulating the 100 hours of pumping system operation. After each two hours of running, the nozzle shall be shut off no less than four times and then fully opened without changing the truck controls. In addition, during each two hours of running, the truck engine shall be shut off for five minutes. There shall be no evidence of leakage, wear, part failure, impending failure or any sort of degradation of the truck as a result of this test.

4.6.9.1 Pumping system operation. After successful completion of all tests in which pumping system operation is mandatory, the total test time of operation shall be recorded. If the recorded time is less than 140 hours, the pumping system shall be operated continuously until the required 140 hours of operation is properly completed.

4.6.10 Hose reel. The servicing hose reel shall demonstrate its ability to meet the requirements of 3.7.21.16 through 3.7.21.16.4.

4.6.11 Flexible coupling hydrostatic tests. The flexible couplings installed in flow lines between the tank and pumping system shall be hydrostatically tested. Each coupling shall be bench tested in a test fixture to simulate the piping on each side of the coupling. The coupling, attachment and clamping arrangement shall be identical to that used on the truck. Flexible couplings on the suction side of the pump shall be tested at 50 PSI or three times the maximum working pressure, whichever is greater, and 15 inches Hg vacuum. Flexible couplings in the tank return line or on the pressure side of the pump shall be tested at 100 PSI or three times the maximum working pressure, whichever is greater. The couplings shall be subjected to 100 maximum pressure cycles (see 6.3) and 100 evacuation cycles (see 6.3) as applicable. There shall be no leaks or evidence of wear, damage or impending failure as a result of this test. It will not be necessary to subject groove type pipe couplings conforming to MIL-C-10387 to this test.

4.6.12 Hydrodynamic loading. The cargo tank shall be filled through the filler opening at a flow rate of at least 200 GPM through a two inch diameter pipe. There shall be no evidence of damage, failure or permanent distortion due to the structure or other characteristics of the design of the tank.

4.6.13 Structural integrity and part removal.

4.6.13.1 Fender test. It shall be demonstrated that each fender will withstand a static load of 250 pounds placed on the top of the fender.

4.6.13.2 Tank inspection platform. It shall be demonstrated that the tank inspection platform can support a load of 350 pounds without permanent deformation.

4.6.13.3 Boarding ladder test. It shall be demonstrated that a 250 pound man can scale the boarding ladder. The ladder shall show no signs of permanent deformation.

4.6.13.4 Pumping compartment roof loading. It shall be demonstrated that the

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pumping compartment roof meets the requirements of 3.7.20.2 without damage to the roof or any portion of the truck.

4.6.14 Electromagnetic interference. The truck shall be tested for radiated interference in accordance with MIL-STD-462 to determine compliance with 3.5.15.

4.6.15 Acoustic noise level. The acoustic noise levels produced by the truck shall not exceed the values specified in 3.7.25 in any octave band at any of the measurement positions specified under 4.6.15.1. All acoustic measurements shall be made during rated capacity operation of the unit. Ambient background noise levels at all measurement positions shall be at least 10dB below the octave band sound pressure levels produced by the truck. The acoustic measurement instrumentation shall be as specified in 4.6.15.2.

4.6.15.1 Measurement position.

4.6.15.1.1 Near field position. Measurements shall be made at seven positions spaced every 45° on a 15 foot radius circle centered on the geometrical center of the truck cab. One of the positions shall be located on the forward longitudinal centerline of the truck. The microphone shall be located five feet above the pavement level and oriented toward the center of the cab.

4.6.15.1.2 Operator's position. Measurements shall be made at the operator's position at the control panel with the microphone five feet above the pavement level.

4.6.15.2 Instrumentation. The basic instrumentation shall consist of a sound level meter which meets the requirements specified in ANSI S1.4. An up to date calibration and frequency response of the entire measurement system, including the microphone, shall be available to permit correction of the measured data. The ambient background sound levels at all measuring positions shall be at least 10dBA below that produced by the truck.

4.6.16 Environmental tests. The truck shall be subjected to the following environmental tests conducted in accordance with the specified method of MIL-STD-810. Unless otherwise specified (see 6.2), and engineering evaluation may be performed in lieu of the testing specified in 4.6.16.4, 4.6.16.5 and 4.6.16.6. The engineering evaluation shall consist of a written detailed analysis of the truck assembly and all its parts and components to satisfactorily accomplish the required environmental conditions.

4.6.16.1 High temperature. The truck shall be subjected to high temperature in accordance with Method 501, Procedure I. The operating temperature at the end of the exposure shall be 125°F or less. The truck shall be filled to capacity with water at a temperature of at least 60°F at the beginning of the 48 hour soak period. At the end of the 48 hour soak period, the truck engine shall be started. The test specified in 4.6.2 and 4.6.6.3 shall be conducted. The truck engine shall be stopped and the pumping system operated for five minutes. The truck engine shall be restarted and operated for a 1/2 hour. During this 1/2 hour period, the pumping system shall be operated to service water at a flow rate of 20 GPM. The servicing operation shall be accomplished by dispensing the water into the tank manhole. The truck engine shall be stopped for five minutes and restarted. All engine starts shall be

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accomplished within five minutes of the original attempt. In addition to the data required by 4.3.4 the following data shall be recorded at two minute intervals during the operational portion of the test:

a. Coolant temperature into engine.

b. Coolant temperature out of engine.

c. Tank water temperature.

d. Engine oil temperature.

4.6.16.2 Low temperature. The truck shall be subjected to low temperature in accordance with Method 502, Procedure I. The truck, completely drained of water, shall be placed in the test chamber. The internal chamber temperature shall be lowered to -65°F and maintained for a period of not less than 48 hours. At the conclusion of the exposure period, the truck shall be removed from the chamber and inspected.

a. The truck shall be returned to the test chamber for 48 hours at -40°F. The cargo tank shall be filled with 250 gallons of water (60°F, plus or minus 5°F) prior to initiation of the -40°F operating test. The 110 volt winterization systems shall be operated throughout the -40°F, 48 hour period. At the end of this 48 hour period, the truck engine shall be started. The truck engine shall start within 15 minutes of the initial attempt without the addition of auxiliary heating or electrical power. The battery shall remain installed in the truck throughout this test, including the soak period.

b. Except as specified in these tests, the heating system (see 3.7.21.33) shall be opeated from the truck 12 volt DC power whenever the truck engine is operating to maintain the pump compartment at least at 55°F. The maximum operating current (amps) of the 12 volt DC pump compartment heating system shall be measured and recorded. The night servicing lights (see 3.7.22.2) shall be turned on whenever the truck engine is operating during this test.

c. After engine warmup, the entire length of the servicing hose shall be extended from the reel. The pumping system shall be operated to service water for at least two minutes at a flow rate of 20 GPM and at least one additional minute at approximately 10 GPM. The servicing operation shall be accomplished by dispensing the water into the tank manhole. The hose shall be immediately rewound on the reel. The reel shall demonstrate proper operation and adequate power under the low temperture conditions. The test of 4.6.2 shall be conducted.

d. The truck engine shall be stopped for five minutes and restarted. This engine start shall be accomplished within 10 minutes or the original attempt. With the pumping system completely filled with water, the pumping compartment heating system shall be shut off for one hour. Following the one hour shutdown and without any heating system operating, the entire length of servicing hose shall be extended from the reel. The pumping system shall be operated to service water for at least two minutes at a flow rate of 20 GPM. The servicing operation shall be accomplished by dispensing the water into the tank manhole. The hose shall be immediately rewound on the reel and the unit inspected.

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e. The pumping compartment heating system (see 3.7.21.33) shall be restarted and an 11 hour soak period shall commence to insure compliance with 3.7.19.3. Following the soak period, the entire length of servicing hose shall be extended from the reel. The pumping system shall be operated to service water for at least two minutes at a flow rate of at least 15 GPM. The servicing operation shall be accomplished by dispensing water into the tank manhole. The hose shall be immediately rewound on the reel and the unit inspected.

4.6.16.3 Operation. The truck shall operate satisfactorily to perform its intended functions throughout this entire test. There shall be no leakage, damage, evidence of degradation or impending failure of any portion of the truck. There shall be no evidence of freezing in any portion of the pumping system, tank and all water bearing components throughout any part of this entire test. In addition to the data required by 4.3.4, the following data shall be recorded at two minute intervals during the operational portion of the tests:

a. Collant temperature into engine.

b. Coolant temperature out of engine.

c. Engine oil temperature.

d. Tank water temperature.

e. Pumping compartment ambient temperature.

4.6.16.4 Humidity. The truck shall be subjected to humidity in accordance with Method 507, Procedure I. A detailed engineering analysis can be supplied in liew of this test.

4.6.16.5 Salt fog. The truck shall be subjected to salt fog in accordance with Method 509, Procedure I. A detailed engineering analysis can be supplied in lieu of this test.

4.6.16.6 Dust. The truck shall be subjected to dust in accordance with Method 510, Procedure I. A detailed engineering analysis can be supplied in lieu of this test.

4.6.17 Air transportability Analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with paragraph 3.5.16. The analysis shall include the tiedowns and all major components and their capability to with stand the g-loadings specified in MIL-STD-1791. The evaluation shall also include a dimensional analysis for the truck while loading aboard a C-130, C-141 or C-5 Aircraft.

4.6.18 Dimensional check. A layout drawing of the truck shall be included in the test report with all dimensions referenced herein measured on the actual vehicle while standing empty on a flat surface.

4.6.19 Weight check. The weight of the truck shall be measured with the unit standing on a flat surface, first empty and then fully loaded with 250 gallons of water, a 250 pound driver and passenger, all standard equipment tools and

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operating manual. The wheel loading requirements of 3.6.5 shall be met.

4.6.20 Mobility tests.

4.6.20.1 Road handling. The truck shall be filled with approximately 200 gallons of water and driven not less than 35 miles over pave highway or similar improved roads and not less than 15 miles over graded gravel or similar unimproved roads (see 6.3). The truck speed shall be 45 MPH to 50 MPH on the improved roads and 25 MPH to 30 MPH on the unimproved roads, wherever practical. At least 10 short radius right turns, 10 short radius left turns and five sudden stops shall be conducted. Throughout this test, the truck shall be observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, etc. Upon completion of this test, the truck shall be inspected for misalignment, distortion or other damage sustained as a result of the testing. Undesirable roadability features and any damage resulting from the test shall be cause for rejection. A complete description of any adverse handling characteristics or damage sustained shall be recorded on the data sheets.

4.6.20.2 High speed. The truck shall be loaded to design capacity. It shall be demonstrated that the truck can achieve a speed of 50 MPH on level paved highway within one mile from a dead stop.

4.6.20.3 Service brakes. The truck shall be loaded to design capacity and subjected to three emergency stops in rapid succession from 20 MPH on clean dry smooth level paved surface free from loose material. Each stop shall be accomplished at no more than one minute intervals. The truck shall stop within 30 feet from the point the brakes are applied. A time distance recorder shall be used to identify the point where the brakes are initially applied.

4.6.20.4 Parking brakes. It shall be demonstrated that the parking brakes will hold the fully loaded truck on a 20 percent slope in both directions.

4.6.20.5 Ramp clearance. The truck in the empty condition shall enter, negotiate and leave a ramp having a slope of 20° in both the forward and reverse directions. There shall be no contact between any part of the truck, except the wheels, with any part of the ramp surface including the horizontal top and bottom landings. The length of the ramp measured along the slope shall be greater than the wheel base of the truck. The bottom and top of the ramp shall end in horizontal landings which form a sharply defined 20° angle with the ramp.

4.6.20.6 Suspension articulation. The fully loaded truck shall be driven over six inch obstacles. The obstacles shall be six inch high blocks, one for each track. The blocks shall first be side by side in each track and then at an angle of 45° across the tracks to simulate the truck negotiating an object at 45° to the path of the truck. The blocks may be sloped to facilitate driving over. There shall be no interference, part failure or permanent or objectionable distortion of any part of the truck as a result of this test.

4.6.20.7 Eight hundred mile test. The truck shall be driven 800 miles consisting of four vehicle endurance cycles of 200 miles. The first cycle shall be with the tank half filled with water, the second and fourth cycle shall be with the tank filled to rated capacity and the third cycle shall be with the tank

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empty. At the end of the second cycle, the pumping system shall be operated 1/2 hour at rated capacity and the hydrostatic test (see 4.6.5) conducted. During the paved highway portion of each vehicle endurance test, at least 25 short radius right turns, 25 short radius left turns and five sudden stops shall be accomplished. At the end of the fourth cycle, the tests specified in 4.6.20.4, 4.6.20.5 and 4.6.20.6 shall be conducted. Each vehicle endurance cycle shall consist of the following phases:

a. Highway - 100 miles at an average speed of at least 40 MPH and a maximum speed of 50 MPH.

b. Unimproved road - 100 miles at an average speed of at least 25 MPH and a maximum speed of 40 MPH.

4.6.20.7.1 Servicing and adjustments. The truck shall be serviced and the brakes adjusted before starting the road testing. Perodic servicing, lubricants and preventive maintenance adjustments will be permitted during the testing at the intervals specified in the operation and service instruction handbook. Tires shall be serviced as necessary. No other servicing, reconditioning or repair will be permitted without prior approval of the procuring activity.

4.6.20.7.2 Inspection. Throughout the road endurance test, the truck shall be observed as to its trackability, ease of handling, tendency to yaw, tendency to sway, etc. Upon completion of the road endurance test, the truck shall be inspected for excessive wear, misalignment, distortion or structural failure or impending failure of any portion of the truck. There shall be no leaks or malfunction to the pumping system. A complete description of any excessive wear, damage, failure or impending failure or roadability deficiencies shall be recorded on the test data sheets.

4.6.21 Reliability demonstration. Qualitative measures of reliability will be demonstrated for both preproduction tests and the operational capability check as follows:

a. Preproduction tests - Satisfactory completion of all tests included in 4.6.1 through 4.6.20 demonstrates compliance with qualitative reliability requirements.

b. Quality conformance inspection - Satisfactory completion of the operational capability check (see 4.5.3) demonstrates compliance with the quantitative reliability requirements. Failure actions, such as rejection of equipment, verification of repairs, restoration of failed equipment and completion of reports shall meet the requirements of MIL-STD-781.

4.6.22 Maintainability demonstration. Compliance with qualitative maintainability requirements shall be demonstrated by inspection or examination of product. Compliance with quantitative maintenance downtime requirements shall be demonstrated by the task performance specified below:

The active maintenance downtime, the tools and equipment required and any obstructions or conditions hindering task

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performance or presenting potential hazard to personnel or parts shall be observed and recorded. Administrative and supply time, such as completing records or obtaining parts, need not count as active downtime. Maintenance task time data from other maintenance and teardown inspections done in connection with other tests specified herein may be applied to the maintainability demonstration if adequate observations are made and recorded. Signs of excessive wear or other indications of potential failure shall be recorded where observed and the information supplied to the design activity for study and recommendation of appropriate modifications to be reflected in items to be delivered. Such recommendations shall be subject to procuring activity approval before being implemented, or successful repeat of other tests specified herein may be required where the test result could be influenced by the modifications. It shall be demonstrated that the equipment is in satisfactory operating condition after the maintenance tasks have been done.

4.6.22.1 Corrective maintenance. Compliance with the corrective maintenance downtime requirements shall be demonstrated by a maintenance demonstration procedure, maintenance task selection and maintenance task performances in accordance with MIL-STD-471 including Appendices A and B, Method 1, Test Plan Pair A1 plus B1. The task selected and done shall be related to the recommended or the stated and assumed maintenance concept.

4.6.22.2 Preventive maintenance. Compliance with the quantitative preventive maintenance downtime requirements shall be demonstrated in accordance with MIL-STD-471, Appendix B, Method 6. Each task need not be done more than once, but the resulting times shall be weighed in calculations to account for the expected frequency of each task in the reference period selected. For maximum preventative maintenance downtime demonstration, less than 0.1 (10 percent) of the task times observed shall require greater downtime than the quantitative requirement. Servicing tasks shall be considered part of, and shall be included in, the demonstration of preventive maintenance.

4.6.22.3 Tools and equipment. Tools and equipment recommended for maintenance shall be used to the maximum practicable extent during testing with emphasis on any special tools or equipment required. Use of special tools and equipment shall be sufficient to determine that they are useful and justified. Instances where special tools or equipment are adequate shall be recorded in detail.

4.7 Refurbishment. After successful completion of all required testing, each vehicle shall be restored to like new condition.

4.8 User test. When specified (see 6.2), the user test sample shall be delivered to a designated site for 45 available days (seven days a week) of user tests. The 45 available days shall be consecutive, except when interrupeted for maintenance/repair downtime, as defined herein. The user test will not begin until after completion of the first article tests, except the environ-

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mental tests, of the contractor tet sample as defined by 4.2.1.1. Run-intest (see 4.4.1) and the examination (see 4.3) shall be performed on the user test sample prior to shipment of the user test sample to the designated site. The user test sample shall not be shipped to the designated site without prior 15 day notification to the PCO. The test period shall begin the day after the vehicle arrives at the designated site and shall end when 45 days of availability are complete or 60 calendar days have elapsed, whichever occurs first. If the user test sample has not been available for 45 test days after 60 calendar days have elapsed, the user test shall be considered a failure. The vehicle shall be available for any one period of eight consecutive hours during each day of the 45 day test. The periods shall be scheduled by the representative of the designated site and the schedules shall be provided to the contractor at least 12 hours in advance. Any period of more than four hours of the same day that the vehicle is out of service for maintenance or repair shall not be considered an available day for the test period. The vehicle shall be operated, and scheduled maintenance shall be performed by Air Force personnel using the operation and maintenance procedures contained in the preliminary technical data to be supplied with the test sample. The preliminary technical data shall contain procedures for preparation/tiedown of the vehicle for shipment. The contractor will provide any special tools/equipment required to perform operators maintenance/scheduled maintenance. Any parts or corrective actions required shall be provided by the contractor. Any maintenance actions performed by the contractor shall not begin before notification to the designated site representative, and Government personnel shall be allowed to observe their actions. The contractor shall provide all vehicle consumables required in the course of the test except for diesel fuel, water, compressed air and electrical power which will be furnished by the designated site. The contractor shall provide on-site familiarization to the operators and mechanics as necessary to perform this test. Test operations will be used in or simulate normal Air Force usage and will not exceed any performance requirements defined by Section 3 of this purchase description. Noncompliance with the performance requirements of this purchase description shall be cause for rejection. The contractor shall maintain a daily log. The log shall be signed by the designated site representative and the contractor after each day of activity. The daily log shall document, but not be limited to the following:

a. Operating time.

b. Tasks completed.

c. Technical manual errors.

d. All failures and corrective actions taken.

e. Operational problems encountered.

f. The designated site representatives comments.

The contractor shall report the user test results, including daily logs of activities, in the first article inspection report.

5 Packaging.

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5.1 Vehicle processing. Preservation, packaging, packing and marking shall be in accordance with the requirements of the contract.

6 Notes.

6.1 Intended use. The A/S32A-24 truck is intended for use in servicing potable or demineralized water to all types of USAF aircraft having such a requirement under worldwide conditions at fixed bases. With the addition of the positive displacement meter (see 3.7.21.15) to the pumping system, the truck is also intended for use in servicing water/alcohol mixtures to certain USAF aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. When specified, a user test will be required (see 4.9).

c. Lettering and numbering to be provided on the truck (see 3.5.25.2).

d. Color of vehicle, if not green (see 3.5.24.2).

e. First article test sample (see 4.4.1).

f. Rustproofing if other than the manufacturer's standard commercial practice is required (see 3.5.24.3).

6.3 Definitions. For the purpose of this specification, the following definitions apply:

6.3.1 Cycle. A cycle is defined as operation of a component from the normal storage position through all of its functions or maximum pressure to its normal storage position.

6.3.2 Failures. A failure is defined as any malfunction which prohibits the vehicle from meeting the performance requirements of Section 3 of the specification.

6.3.3 Servicing system. The servicing system is the servicing hose reel, servicing hose, nozzle assembly, valving and associated equipment.

6.3.4 Self priming centrifugal pump. A self priming centrifugal pump is one designed to have the ability to prime itself automatically after being initially filled when operating under a suction lift, to free itself of entrained gas without losing its prime and to continue normal pumping without attention.

6.3.5 Unimproved road. An unimproved road is a road consisting of level to level gravel trails.

6.4 Subject term (key word) listing.

a. Truck, Tank

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b. Engine, Diesel

c. Automatic transmission

d. Air Transportable

e. Pumping system
f. Potable water

Certain provisions of this specification in 3.7.21.17.5 are the subject or international standardization agreement ASCC Air Standard 17/34. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

Custodian
Air Force - 99

Preparing Activity
Air Force - 84
Agent: Air Force - 99
Project Number:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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